

# Supplementary Materials for: Predicting the long-term impact of AI assistants on collective intelligence

Jaeseung Kim<sup>1</sup> Sang Woo Park<sup>1,2,\*</sup>

<sup>1</sup> School of Biological Sciences, Seoul National University, Seoul, Korea

<sup>2</sup> Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

\*Corresponding author: sangwoopark@snu.ac.kr

## Materials and Methods

### Prediction target

Following previous models of CI, we model the prediction target as a linear combination of  $M$  independent causal factors (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (\text{S1})$$

where the coefficient  $\alpha_i$  determines the magnitude and direction of the effect of factor  $i$ , and  $\alpha_0$  represents a global intercept. Each causal factor  $X_i$  is modeled as an independent Gaussian random variable with mean zero and standard deviation  $\sigma_i$ .

### Players

We consider a well-mixed population  $\mathcal{N}$  consisting of  $N$  players engaged in a prediction task (Wang et al., 2025). In the absence of AI assistance, the strategy of player  $A$  is characterized by two variables: interest  $e_A$  and belief  $c_A$ . Player  $A$  observes the causal factor  $X_{e_A}$  and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (\text{S2})$$

where  $e_A$  can take an integer value from 0 to  $M$ . When  $e_A = 0$ , the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that  $X_0 = 1$ .

When the group is assisted by AI, each player's strategy additionally includes reliance on AI, denoted by  $\beta_A$ . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (\text{S3})$$

where  $\beta_A$  ranges from 0 to 1. A value of  $\beta_A = 1$  indicates complete reliance on AI, whereas  $\beta_A = 0$  indicates that the player does not use the AI prediction.

## AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor  $i$  is denoted by  $b_i$ , and the stochastic error is denoted by  $\epsilon$ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume  $b_i = b$  for the main analysis for all  $i$ . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation  $\tau$ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold  $\tau$  constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all  $M$  causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player  $A$  is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (\text{S4})$$

The second is termed the Chatbot AI, which can use only the single causal factor  $X_{e_A}$  associated with the player's interest. The prediction generated by the Chatbot AI for player  $A$  is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (\text{S5})$$

## Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous work (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all  $M$  factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (\text{S6})$$

Second, when the group is assisted by the Chatbot AI, both the player’s initial prediction and the AI’s prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor’s contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (\text{S7})$$

where  $\mathcal{A}_i$  represents the set of players interested in factor  $i$ , and  $n_i = |\mathcal{A}_i|$  represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

## Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player  $A$ ’s revised prediction  $Y_A^{\text{re}}$  and the collective prediction  $\hat{Y}^{\text{re}}$ , the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (\text{S8})$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player  $A$  is weighted by  $\rho_{e_A} = n_{e_A}/N$ , the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (\text{S9})$$

Previous research has shown that these two incentive structures can produce near perfect collective accuracy through social learning, except when the niche-expert incentive is combined with simple averaging (Wang et al., 2025). They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

99 To examine the effects of directly encouraging or discouraging AI use, we mod-  
 100 ify the feedback and niche-expert structures by adding a term proportional to the  
 101 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (\text{S10})$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (\text{S11})$$

102 Here,  $\lambda$  represents the strength of the direct incentive or penalty applied to AI use. A  
 103 positive value of  $\lambda$  rewards AI reliance, whereas a negative value penalizes it. When  
 104  $\lambda = 0$ , no direct incentive or penalty is applied, and the payoff functions reduce to  
 105 the original feedback and niche-expert structures.

106 The balanced structure is defined as the weighted mean of the feedback and  
 107 niche-expert payoff components, with balancing weight  $w$ :

$$\pi_A = (1 - w)Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - w\rho_{e_A}(Y_A^{\text{re}} - Y)^2, \quad (\text{S12})$$

108 where  $w \in [0, 1]$  determines the relative contribution of the feedback and niche-expert  
 109 components. We set  $w = 0.5$  for all analyses unless otherwise specified, such that  
 110 the two components contribute equally to the balanced structure. The sensitivity of  
 111 the results to  $w$  is examined in Supplementary Fig. S7.

112 To examine whether incentives based on unrevised predictions (i.e., predictions  
 113 made without incorporating AI predictions) could overcome the limitations of the  
 114 other incentive structures, we defined an unrevised balanced structure. The unre-  
 115 vised balanced structure is defined as the arithmetic mean of the feedback structure  
 116 based on the unrevised collective prediction and the niche-expert structure based on  
 117 unrevised individual predictions:

$$\pi_A = \frac{Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2}{2}, \quad (\text{S13})$$

118 where  $\hat{Y}$  is the collective prediction based on unrevised individual predictions, and  
 119  $Y_A(Y - \hat{Y})$  and  $-\rho_{e_A}(Y_A - Y)^2$  represent the unrevised feedback and unrevised niche-  
 120 expert components, respectively.

121 We further defined a combined balanced structure as the arithmetic mean of  
 122 the balanced and unrevised balanced structures, with the balancing weight set to  
 123  $w = 0.5$ :

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2 + Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2}{4}. \quad (\text{S14})$$

## 124 Simulations

125 All simulations are conducted using an agent-based model with 50 causal factors  
 126 ( $M = 50$ ) and 10,000 players ( $N = 10,000$ ). The parameters defining the prediction  
 127 target are sampled at the beginning of the simulations and subsequently held fixed.

Coefficients  $\alpha_i$  are sampled from a uniform distribution on  $[-5, 5]$ , except in the simulation presented in Figure 4, in which  $\alpha_i$  is set to  $5 - 0.2i$  for visualization. Standard deviations  $\sigma_i$  are sampled from a uniform distribution on  $[0, 3]$ .

Players’ initial strategies are also sampled from predefined probability distributions. The distribution of individual belief  $c_A$  depends on the aggregation rule. Under simple averaging, each factor’s contribution is scaled by the proportion of players interested in that factor. We therefore sample beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of  $c_A$  is required. The initial interest of each player is sampled uniformly from the integers between 0 and 50, and reliance on AI is sampled from a uniform distribution on  $[0, 1]$ . To account for stochastic variation, we perform 30 independent simulation replicates with randomly initialized player populations for each condition presented in the main figures.

We assume that prediction events occur on a substantially faster timescale than social learning. In each generation, we therefore calculate the expected payoff  $\mathbb{E}[\pi_A]$  of each player analytically rather than explicitly sampling prediction events. The expectation is taken over the distributions of the causal factors  $X_1, \dots, X_M$  and the stochastic AI error  $\epsilon$ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs are derived for all incentive structures and are presented in the Supplementary Text.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players,  $A$  and  $B$ , are randomly sampled, and player  $A$  imitated all three strategies of player  $B$  with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (\text{S15})$$

where  $s$  represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate  $\mu$ , such that with probability  $\mu$ , player  $A$  does not imitate player  $B$  but instead redraw all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4.

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

169 Collective accuracy is normalized relative to the baseline prediction  $\alpha_0$ . A nega-  
170 tive value indicates that the collective prediction performs worse than the baseline.  
171 Counterfactual collective accuracy is evaluated by setting  $\beta_A = 0$  while retaining the  
172 evolved interests and beliefs of the AI-assisted population. In calculating interest  
173 diversity, we use the convention that  $\rho_i \ln \rho_i = 0$  when  $\rho_i = 0$  (Table S1).

## Supplementary Text

Here, we derive the expected payoffs and collective accuracy used in the simulations. We consider Omniscient AI and Chatbot AI separately because they contain different amounts of information and therefore require different aggregation rules. We then analyze the corresponding evolutionary dynamics under a deterministic large-population approximation.

### Omniscient AI

We first consider Omniscient AI, which incorporates all causal factors regardless of the player's interest. We derive individual and collective predictions before calculating the expected payoffs under each incentive structure.

#### Model

Because Omniscient AI observes all  $M$  causal factors, its prediction for player  $A$  is given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S16})$$

where  $b_i$  represent factor-specific biases and  $\epsilon$  represents stochastic error, following a Gaussian distribution with mean zero and standard deviation  $\tau$ . A new stochastic error is generated independently for each player and prediction event.

Let  $B = \sum_{i=0}^M b_i$  denote the total additive bias. Because the true prediction target is  $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$ , the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S17})$$

In contrast to the AI prediction, player  $A$ 's initial prediction depends only on the causal factor associated with their interest. Their initial and revised predictions are therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S18})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S19})$$

represents player  $A$ 's interest,  $c_A$  represents their belief about the effect of the corresponding factor, and  $\beta_A$  represents their reliance on AI. As in the main text, we set  $X_0 = 1$ , such that a player with  $e_A = 0$  makes a constant initial prediction.

## 198 Aggregation

199 Because the AI component of each revised prediction contains information about all  
200  $M$  causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (\text{S20})$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (\text{S21})$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon_j) + (1 - \beta_j) c_j X_{e_j}\} \quad (\text{S22})$$

201 To separate the contributions of AI and the players' initial predictions, we define the  
202 population's mean reliance on AI, the stochastic AI error remaining after aggregation,  
203 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (\text{S23})$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon_j \quad (\text{S24})$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S25})$$

204 Here,  $\bar{\beta}$  represents the mean reliance on AI,  $\bar{\eta}$  represents the stochastic AI error  
205 remaining after averaging, and  $\gamma_i$  represents the contribution of players' initial pre-  
206 dictions to factor  $i$ . Using these definitions, the revised collective prediction can be  
207 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S26})$$

208 Thus, the collective prediction consists of an AI-derived component, the contributions  
209 of the players' initial predictions, and the stochastic AI error that remains after  
210 aggregation.

## 211 Expected payoffs

212 We next derive the expected payoff under each incentive structure. Expectations are  
213 taken over the distributions of the causal factors and stochastic AI errors, conditional  
214 on the current population strategies and the fixed parameters of the prediction target.



## 215 Feedback

216 Under the feedback structure, a player's payoff depends on the alignment between  
 217 their revised prediction and the residual error of the collective prediction. To express  
 218 this residual, we first define the remaining error in the intercept and in the coefficient  
 219 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S27})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S28})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S29})$$

220 where  $\delta_0$  represents the remaining error in the intercept, whereas  $\delta_i$  represents the  
 221 remaining error associated with factor  $i$ . Because the causal factors and stochastic AI  
 222 errors are independent and have zero mean, cross terms vanish when the expectation  
 223 is taken. Substituting the revised individual prediction and collective residual into  
 224 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S30})$$

$$= \mathbb{E} \left[ \left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S31})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \quad (\text{S32})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S33})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S34})$$

225 The first term represents the alignment between the AI-derived prediction and the  
 226 collective residual, whereas the second term represents the contribution of the player's  
 227 initial prediction. The final term arises because the stochastic error in player  $A$ 's AI  
 228 prediction also contributes to the collective prediction. This finite-population effect  
 229 decreases as population size increases.

## 230 Niche-expert

231 Under the niche-expert structure, the payoff is the negative squared error of the  
 232 revised individual prediction, weighted by the frequency of the player's interest.  
 233 Because  $X_0 = 1$ , a player with  $e_A = 0$  makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S35})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S36})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S37})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S38})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S39})$$

For a player with  $e_A \neq 0$ , the initial prediction instead depends on the observed value of factor  $X_{e_A}$ . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S40})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S41})$$

$$= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B\right. \quad (\text{S42})$$

$$\left. + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S43})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S44})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2 \quad (\text{S45})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S46})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

## Incentivizing or penalizing AI use

Direct incentives add the term  $\lambda\beta_A$  to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply  $\lambda\beta_A$ . The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A, \quad (\text{S47})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = \mathbb{E}\left[-\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S48})$$

Thus, a positive value of  $\lambda$  increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

## 247 Expected accuracy

248 Using the residual defined above, the mean squared error of the revised collective  
249 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[ \left( \alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S49})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[ \left( \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S50})$$

250 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y}^{re})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S51})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S52})$$

251 The collective prediction error consists of the squared error in the intercept, the re-  
252 maining error associated with each causal factor, and the stochastic AI error remain-  
253 ing after averaging across players. Counterfactual collective accuracy is obtained  
254 from the same expression by setting  $\beta_A = 0$  for all players while retaining their  
255 evolved interests and beliefs.

## 256 Chatbot AI

257 Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each  
258 player's interest. This causes the information contained in AI predictions to depend  
259 directly on the distribution of interests in the population. Here, we derive the collec-  
260 tive prediction, expected payoffs, and expected collective accuracy under this form  
261 of AI assistance.

## 262 Model

263 The prediction generated by Chatbot AI for player  $A$  depends on the causal factor  
264 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S53})$$

265 where  $b_{e_A}$  represents factor-specific additive bias, and  $\epsilon$  represents stochastic error.  
266 The stochastic error follows a Gaussian distribution with mean zero and standard  
267 deviation  $\tau$  and is generated independently across players and prediction events. The

268 player's initial and revised predictions are given by Player A's initial and revised  
 269 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S54})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S55})$$

270 Here,  $c_A$  represents player A's belief about the effect of their focal factor,  $e_A$  repre-  
 271 sents their interest, and  $\beta_A$  represents their reliance on AI. Thus, the revised predic-  
 272 tion remains specific to the single factor associated with the player's interest.

## 273 Aggregation

274 Because both the player's initial prediction and the Chatbot AI prediction contain  
 275 information about the same causal factor, we use clustering aggregation. Revised  
 276 predictions are first averaged among players interested in the same factor and then  
 277 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S56})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S57})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [ \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon_A ], \quad (\text{S58})$$

278 where  $\mathcal{A}_i$  is the set of players interested in factor  $i$ , and  $n_i = |\mathcal{A}_i|$  is the number of  
 279 players in that set.

280 To simplify the collective prediction, we define the effective coefficient, weighted  
 281 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S59})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S60})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S61})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon_A, \quad (\text{S62})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S63})$$

where  $\tilde{c}_i$  represents the mean effective coefficient for factor  $i$  after AI assistance, whereas  $\tilde{B}_i$  and  $\tilde{\eta}_i$  represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S64})$$

where  $\mathcal{I} = \{i \in \{0, \dots, M\} > 0\}$  denotes the set of represented interests.

## Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

## Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S65})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S66})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S67})$$

where  $\delta_0$  includes the remaining error in the intercept and the total additive AI bias, whereas  $\delta_i$  measures the remaining error in the coefficient of factor  $i$ . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S68})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S69})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S70})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S71})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

## 302 Niche-expert

303 Under the niche-expert structure, the negative squared error of each revised indi-  
 304 vidual prediction is weighted by the frequency of the player's interest. We consider  
 305  $e_A = 0$  separately because these players make constant predictions rather than factor-  
 306 dependent predictions. When  $e_A = 0$ , the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S72})$$

$$= -\mathbb{E} \left[ \rho_0 \left[ \left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S73})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S74})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S75})$$

307 When  $e_A \neq 0$ , the player's prediction varies with their focal causal factor. The  
 308 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S76})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S77})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S78})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S79})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S80})$$

309 In both cases, the expected squared error consists of the mismatch between the  
 310 player's belief and the true coefficient, contributions from causal factors omitted  
 311 from the individual prediction, factor-specific AI bias, and stochastic AI error. The  
 312 niche-expert weight  $\rho_{e_A}$  scales these errors according to the frequency of the player's  
 313 interest.

## 314 Incentivizing or penalizing AI use

315 Directly incentivizing or penalizing AI use adds the term  $\lambda \beta_A$  to the player's payoff.  
 316 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[ Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S81})$$

$$= \mathbb{E} \left[ Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S82})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = \mathbb{E} \left[ -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S83})$$

$$= -\mathbb{E} \left[ \rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S84})$$

### Balanced

Under the balanced structure, the payoff corresponds to the weighted mean of the feedback and niche-expert payoff components, with balancing weight  $w$ :

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[ (1-w)Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - w\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] \quad (\text{S85})$$

$$= (1-w)\mathbb{E} \left[ Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - w\mathbb{E} \left[ \rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] \quad (\text{S86})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

### Triple balanced

### Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept  $\alpha_0$  serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[ \left( \alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S87})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] = \mathbb{E} \left[ \left( \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S88})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

334 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y}^{re})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S89})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S90})$$

335 Here,  $R = 1$  represents perfect collective prediction,  $R = 0$  represents performance  
336 equal to the baseline, and  $R < 0$  represents performance worse than the baseline.

## 337 Evolutionary dynamics

338 Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al.,  
339 2005), we use replicator dynamics as a deterministic large-population approximation  
340 to characterize equilibrium states and their local stability. The agent-based sim-  
341 ulations themselves follow the Fermi imitation rule defined above. At equal-payoff  
342 stationary states, the replicator and Fermi formulations imply the same local invasion  
343 directions for any finite selection strength  $s > 0$ , although their transient dynamics  
344 need not coincide. We focus on the feedback and niche-expert structures, for which  
345 the long-term dynamics can be characterized analytically. The direct interventions  
346 on AI use and the balanced structure are examined through simulations.

347 Let  $\lambda$  denote a strategy,  $p_\lambda$  its frequency, and  $u_\lambda = \mathbb{E}(\pi_\lambda)$  its expected payoff.  
348 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S91})$$

349 where  $\bar{u} = \sum_\lambda p_\lambda u_\lambda$  represents the mean expected payoff of the population. At an  
350 internal stationary state, all strategies with positive frequency must have the same  
351 expected payoff. A boundary state can also be stationary because a strategy with zero  
352 frequency cannot increase unless it is introduced into the population. We therefore  
353 assess the stability of boundary states by asking whether they can be invaded by a  
354 rare alternative strategy.

## 355 Omniscient AI

356 With Omniscient AI, every AI prediction contains information about the full predic-  
357 tion target, and revised predictions are combined through averaging. As population  
358 size increases, independent AI errors vanish from the collective prediction. How-  
359 ever, these errors remain in individual prediction errors and can therefore continue  
360 to affect niche-expert payoffs. We consider the two incentive structures separately.



## 361 Feedback

362 Under the feedback structure, both the stochastic error in the collective prediction,  
 363  $\bar{\eta}$ , and its contribution to expected payoff vanish as  $N \rightarrow \infty$ . Following previous  
 364 work (Wang et al., 2025), we define the Lyapunov function as the expected error of  
 365 collective prediction:

$$V = \mathbb{E} \left[ (Y - \hat{Y}^{re})^2 \right]. \quad (\text{S92})$$

366 The function  $V$  is non-negative and equals zero when the collective prediction is  
 367 identical to the target. To determine how  $V$  changes over time, we take its time  
 368 derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[ (Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S93})$$

$$= -2\mathbb{E} \left[ \left( \delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S94})$$

$$= -2 \left[ \left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S95})$$

369 For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S96})$$

370 Although  $K$  changes with the population state, it is common to all strategies at a  
 371 given state and can therefore be treated as a constant with respect to the strategy  
 372 index  $\lambda$ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S97})$$

373 Under the replicator equation, the rate of change in the population mean of a  
 374 strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S98})$$

375 Similarly, the terms involving  $\gamma_i$  can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S99})$$

376 Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S100})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S101})$$

377 For the feedback structure in the limit  $N \rightarrow \infty$ , the expected payoff of strategy  
 378  $\lambda$  is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S102})$$

379 This expected payoff is identical to the quantity inside the covariance above. There-  
 380 fore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S103})$$

381 Therefore, the expected error of collective prediction cannot increase during the  
 382 evolutionary process and decreases whenever strategies differ in their expected pay-  
 383 offs. The derivative becomes zero when the variance of expected payoffs is zero,  
 384  $\text{Var}(u) = 0$ . This condition arises when all represented strategies receive the same  
 385 expected payoff. However, this condition does not necessarily imply that  $V = 0$ ,  
 386 because a population can lack strategies that would improve its collective prediction.

387 If  $V > 0$ , at least one component of the collective residual remains nonzero. A rare  
 388 strategy whose prediction is aligned with this residual can obtain a higher expected  
 389 payoff than the resident strategies and invade the population. Thus, provided that  
 390 novel strategies can be introduced, an evolutionarily stable state must satisfy  $V = 0$ .  
 391 At such a state,  $\hat{Y}^{\text{re}} = Y$ , and collective accuracy reaches  $R = 1$ .

## 392 Niche-expert

393 Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each  
 394 individual. Stochastic AI error therefore remains in the expected payoff even when  
 395  $N \rightarrow \infty$ , and we retain  $\tau > 0$  throughout this analysis.

396 We first consider strategies having the same interest by optimizing the expected  
 397 payoff with respect to  $c$  and  $\beta$ . For a nonzero interest  $e$ , the expected payoff is strictly  
 398 concave with respect to  $c$  for  $\beta < 1$  and  $\sigma_e > 0$ , and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S104})$$

399 After substituting  $c_e^* = \alpha_e$ , the payoff is strictly concave with respect to  $\beta$ . Its  
 400 unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S105})$$

401 Because  $\beta$  is constrained between 0 and 1, the optimal  $\beta_e^*$  is  $\tilde{\beta}_e$  when  $0 < \tilde{\beta}_e < 1$ ,  
 402 and otherwise lies at the corresponding boundary,  $\beta_e^* = 0$  or  $\beta_e^* = 1$ . Thus, the  
 403 boundary cases do not require a separate evolutionary argument; they only change  
 404 the optimal prediction and its minimum expected squared error. When  $\beta_e^* = 1$ , the  
 405 revised prediction is independent of  $c_e$ , so different values of  $c_e$  are selectively neutral  
 406 and can be treated as a single payoff-equivalent strategy class.

407 To compare the optimized strategies across interests, let  $L_e$  denote the minimum  
 408 expected individual squared error within interest  $e$ ,  $\mathbb{E}[(Y_A^{re} - Y)^2]$ . For an internal  
 409 optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left( \alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S106})$$

410 If the optimum is located at  $\beta_e^* = 0$  or  $\beta_e^* = 1$ ,  $L_e$  is instead obtained by evaluating  
 411 the expected squared prediction error at the corresponding boundary.

412 Interest 0 must be considered separately because  $X_0 = 1$  and the player's belief  
 413 determines a constant prediction. Optimizing with respect to  $c$  gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S107})$$

414 Substituting this into the expected payoff gives a strictly concave function of  $\beta$ , with  
 415 the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S108})$$

416 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S109})$$

417 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S110})$$

418 At this optimum, the player adjusts their belief to offset the systematic AI bias. The  
 419 stochastic component of AI error cannot be removed in the same way and therefore  
 420 limits the benefit of AI reliance.

421 We denote the optimal strategy within interest  $e$  by  $S_e$ . When  $\beta_e^* = 1$ , different  
 422 values of  $c_e$  are selectively neutral and can be treated as a single payoff-equivalent  
 423 strategy class. A stable state can contain only the optimal strategy class within each  
 424 represented interest because a suboptimal strategy can be invaded by  $S_e$ .

425 We next consider the frequencies of the  $M + 1$  optimal strategies. Let  $\rho_e$  denote  
 426 the frequency of  $S_e$ . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S111})$$

427 Assuming  $L_e > 0$  for all interests, the equal-payoff condition of the replicator equa-  
 428 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S112})$$

429 Together with  $\sum_{e=0}^M \rho_e^* = 1$ , this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left( \sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S113})$$

430 This internal stationary state is locally stable. At equilibrium, all represented in-  
 431 terests have the same expected payoff, with  $\rho_e^* L_e = D$ . If the frequency of one  
 432 interest increases slightly above its equilibrium value, its payoff decreases because  
 433 the niche-expert payoff is proportional to  $-\rho_e L_e$ . Conversely, if the frequency of an  
 434 interest decreases below its equilibrium value, its payoff increases. The replicator dy-  
 435 namics therefore generate a restoring force that reduces small deviations in interest  
 436 frequencies and drives the population back toward the internal stationary state.

437 If one or more interests are absent, the resulting boundary state can be stationary  
 438 under the replicator equation, but it is unstable against invasion. The payoff of a rare  
 439 absent interest approaches zero as  $\rho_e \rightarrow 0$ , whereas the mean payoff of the resident  
 440 population is negative when  $L_e > 0$ , so the rare interest increases when introduced  
 441 at sufficiently small frequency.

442 Thus, there exists a unique stable full-support equilibrium in terms of the optimal  
 443 prediction strategies and their frequencies. Whether the constrained optimum has  
 444  $\beta_e^* = 0$ ,  $0 < \beta_e^* < 1$ , or  $\beta_e^* = 1$  does not change the frequency argument, because  
 445 these cases differ only through the resulting value of  $L_e$ . When  $\beta_e^* = 1$ , the strategy  
 446 is selectively neutral with respect to  $c_e$ , but its prediction, payoff, and equilibrium  
 447 interest frequency remain uniquely determined.

448 Having characterized this equilibrium, we next compare its collective prediction  
 449 with a prediction made entirely by AI. In particular, the expected squared error of  
 450 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S114})$$

451 For every interest, complete reliance on AI,  $\beta = 1$ , is a feasible strategy. There-  
 452 fore, because  $L_e$  is the minimum individual expected squared prediction error within  
 453 interest  $e$ ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S115})$$

454 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S116})$$

455 Because the stable equilibrium has  $\rho_0^* > 0$ ,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S117})$$

456 The collective prediction is the average of individual revised predictions. By convex-  
 457 ity of the squared error,

$$\mathbb{E} \left[ (Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S118})$$

458 Therefore,

$$\mathbb{E} \left[ (Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[ (Y - Y^{\text{AI}})^2 \right]. \quad (\text{S119})$$

459 Thus, the diversity of optimized predictions allows the population to outperform an  
 460 individual AI prediction at equilibrium.

461 However, outperforming AI does not imply perfect collective accuracy. Under  
 462 averaging, the non-AI contribution associated with a particular factor is provided  
 463 only by individuals interested in that factor. The predictions of individuals with  
 464 other interests therefore attenuate that factor's coefficient in the collective prediction.  
 465 For a nonzero factor  $i$ , the residual coefficient of the stationary collective prediction  
 466 can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S120})$$

467 At the stable equilibrium,  $\rho_0^* > 0$ . In addition,  $\tau > 0$  implies  $\beta_0^* < 1$ . Therefore, for  
 468 every factor with  $\alpha_i \neq 0$ ,

$$\delta_i^* \neq 0. \quad (\text{S121})$$

469 Because  $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$ , at least one factor contributes positive prediction error.  
 470 Hence,

$$\mathbb{E} \left[ (Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S122})$$

471 and consequently the stationary collective accuracy satisfies  $R^* < 1$ . Thus, the niche-  
 472 expert structure preserves interest diversity and allows the population to outperform  
 473 AI, but averaging prevents the population from reaching perfect collective accuracy.

## 474 Incentive structure incorproating unrevised predic- 475 tions

## 476 Different

## Supplementary Figure

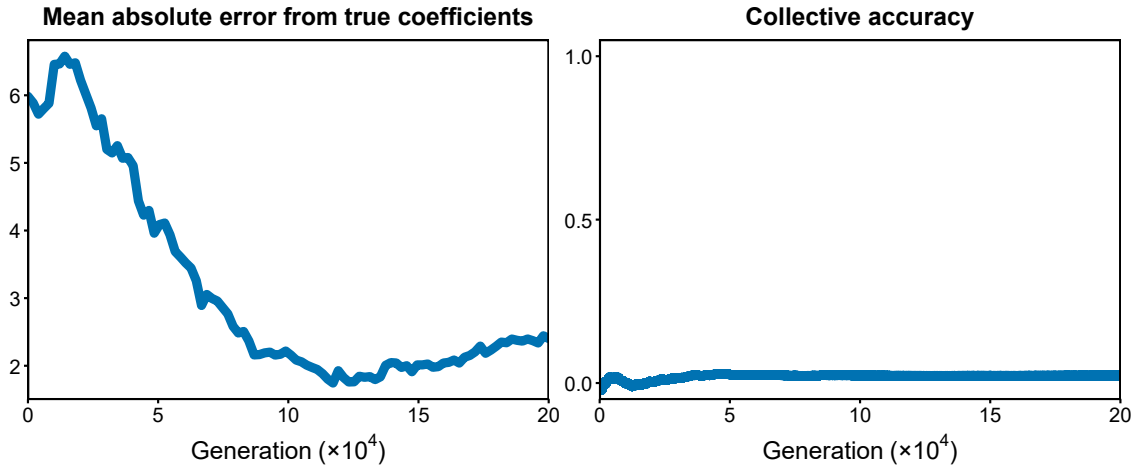


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all  $M + 1$  interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 100^2)$ .

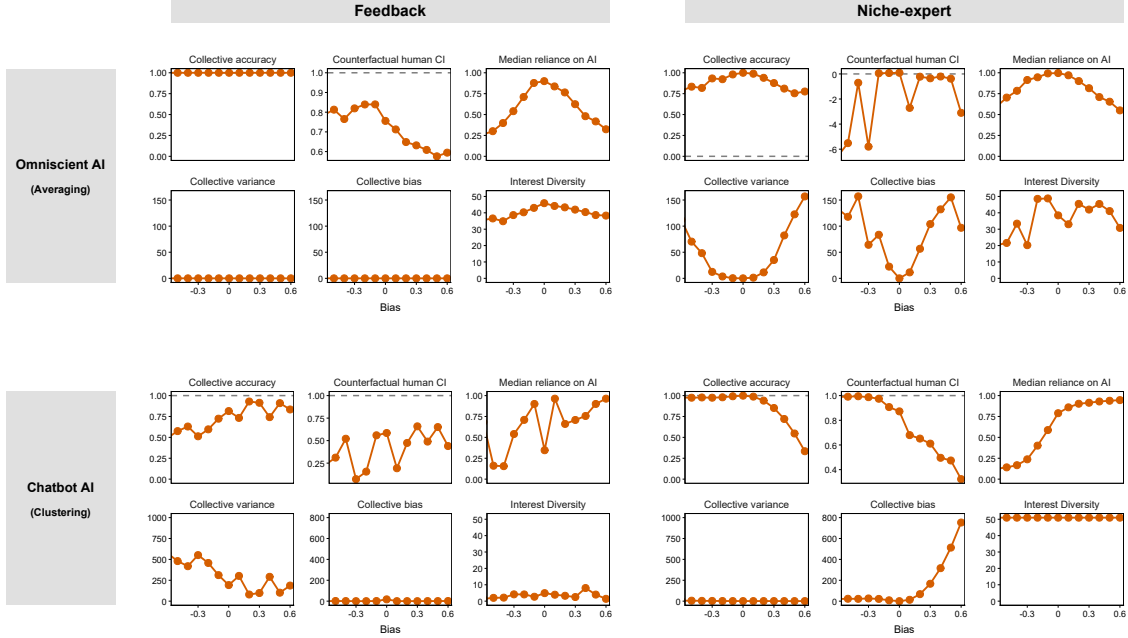


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI ( $\beta$ ). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 100^2)$  for (A, B),  $c_A \sim \mathcal{N}(0, 5^2)$  for (C, D), and  $\beta_A \sim U(0, 1)$ . AI parameters:  $\tau = 0.3$ .

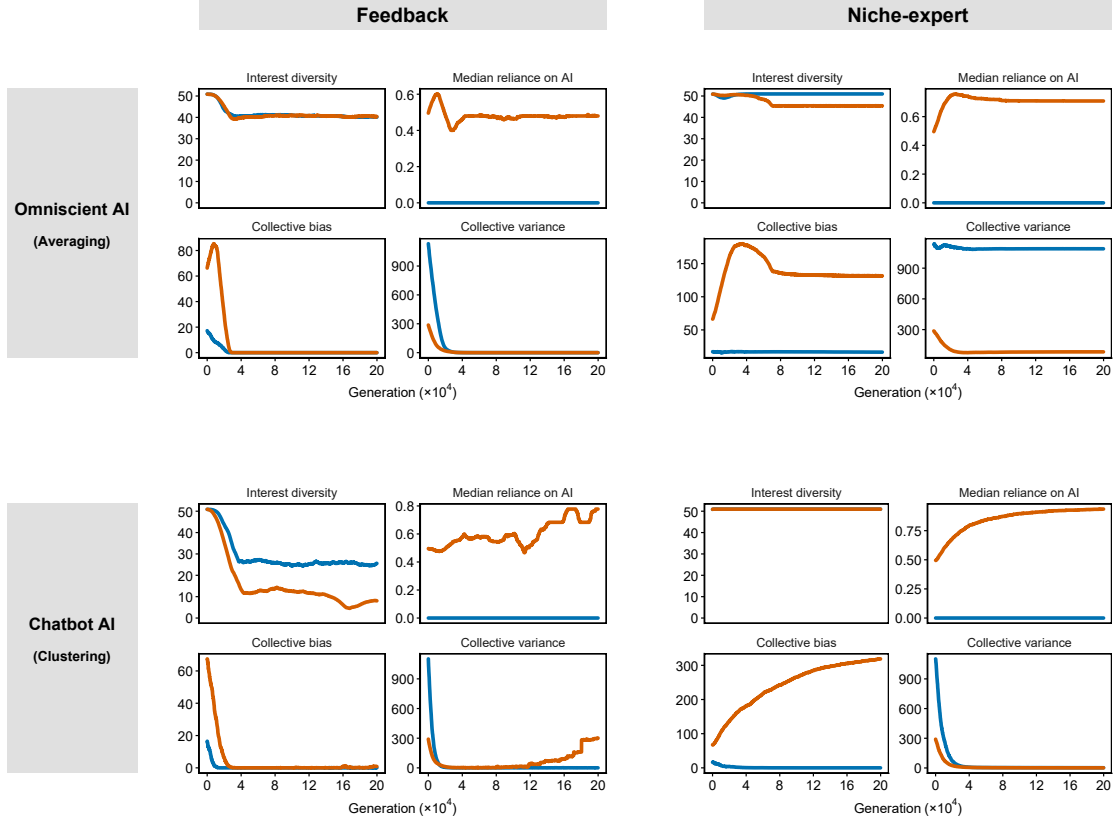


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as Hill-Shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI ( $\beta$ ), collective bias represent systematic deviation of the collective prediction from the true target, collective variance represent variability of the collective prediction error across prediction events. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 100^2)$  for (A, B),  $c_A \sim \mathcal{N}(0, 5^2)$  for (C, D), and  $\beta_A \sim U(0, 1)$ . AI parameters:  $b_i = 0.4$  and  $\tau = 0.3$ .



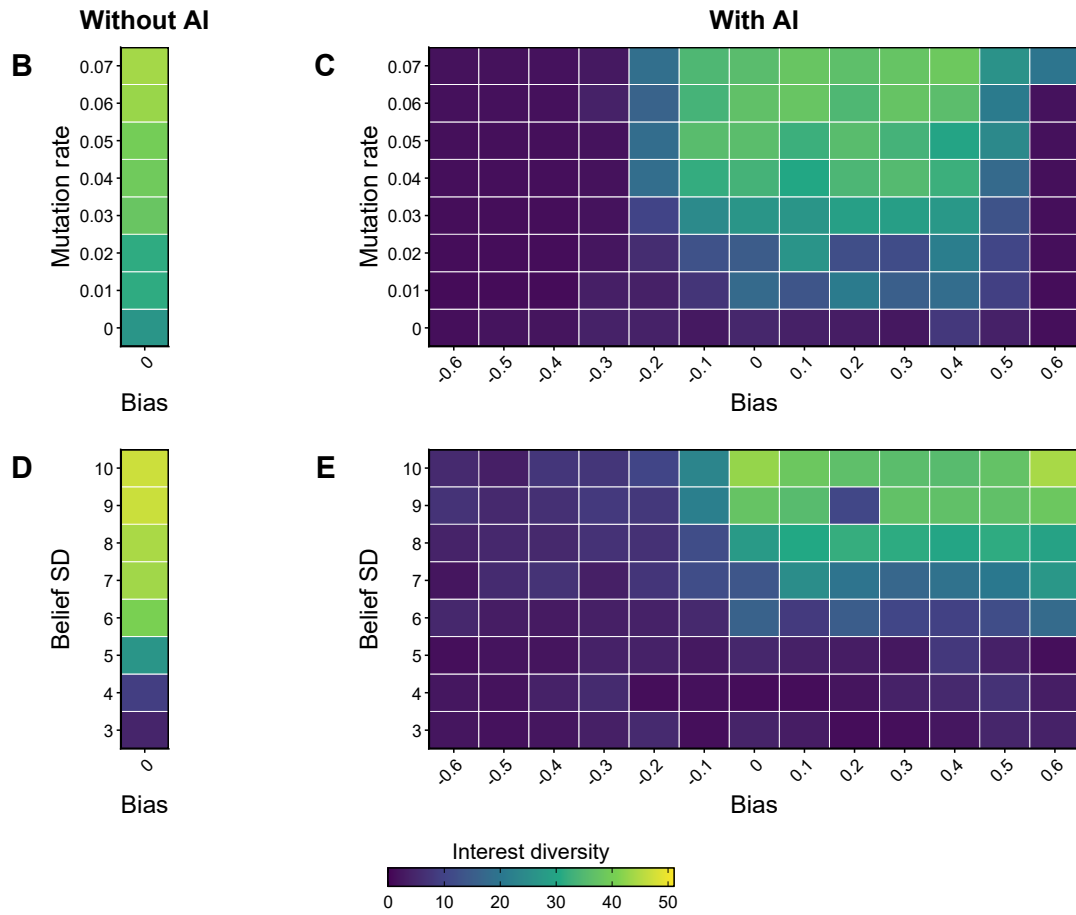
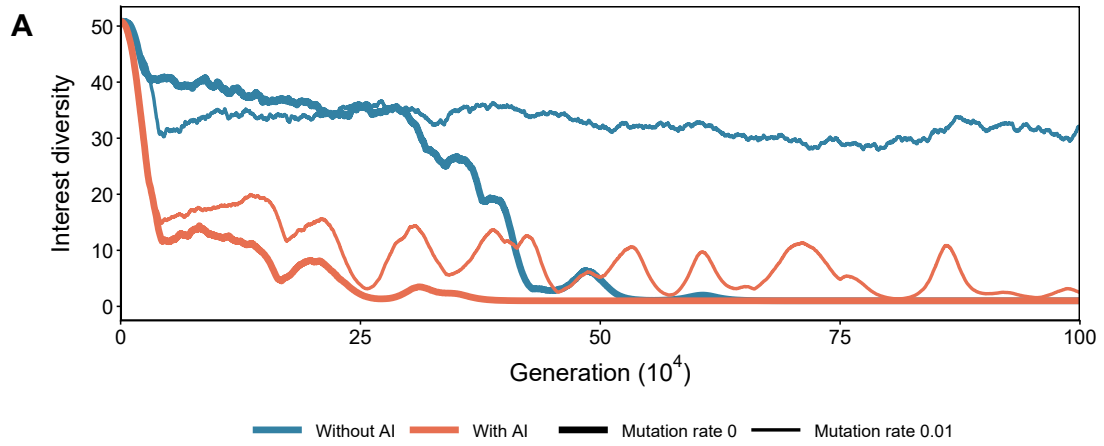


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,001–200,000. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$  for (D, E),  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 5^2)$  for (A, B, C), and  $\beta_A \sim U(0, 1)$ . AI parameters:  $b_i = 0.4$  and  $\tau = 0.3$ , Chatbot AI is used.

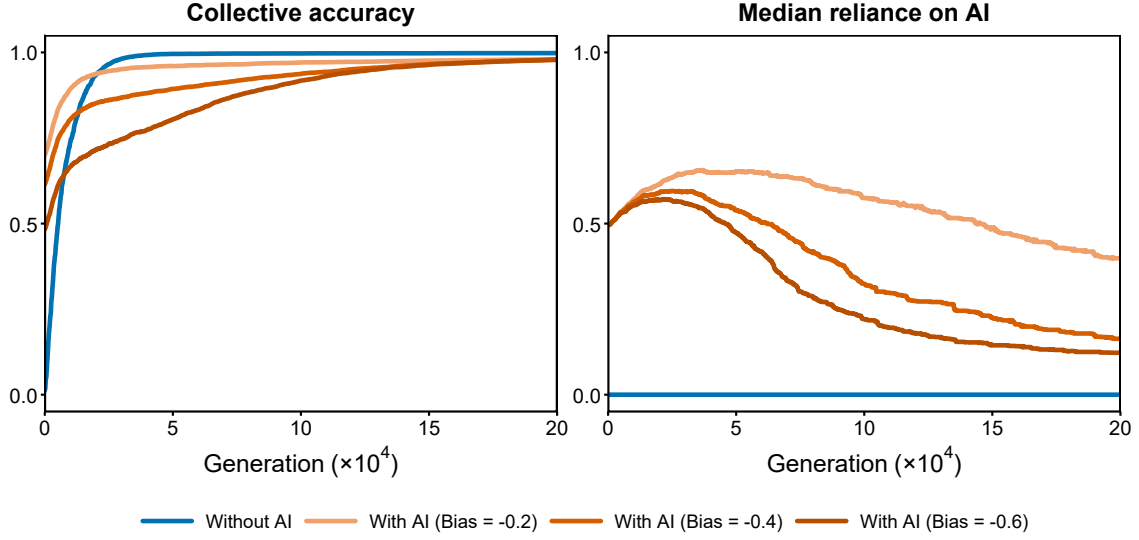


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the  $\alpha_0$  and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 5^2)$ , and  $\beta_A \sim U(0, 1)$ . AI parameters:  $\tau = 0.3$ , Chatbot AI is used.

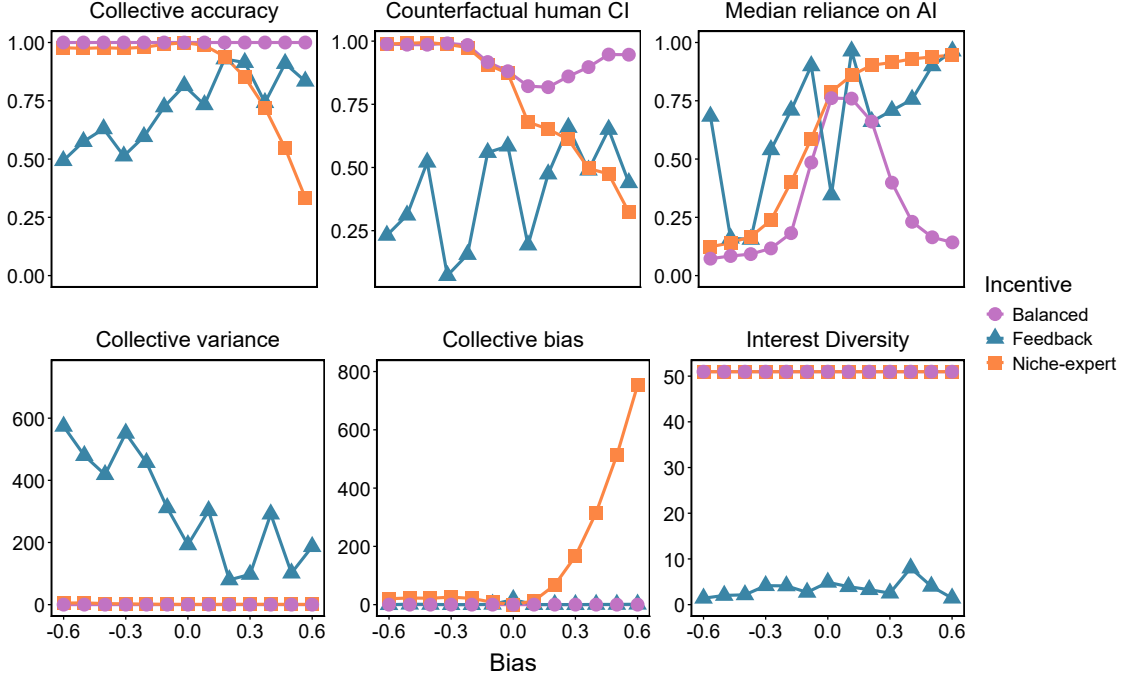


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,001–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 5^2)$ , and  $\beta_A \sim U(0, 1)$ . AI parameters:  $\tau = 0.3$ , Chatbot AI is used.

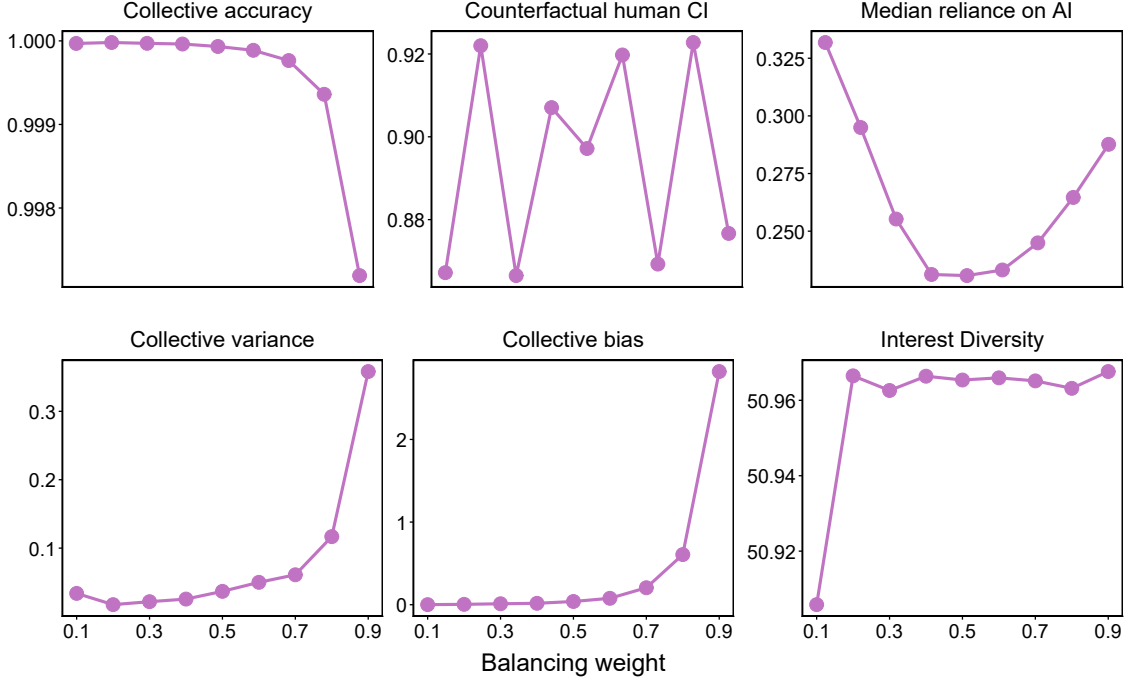


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 5^2)$ , and  $\beta_A \sim U(0, 1)$ . AI parameters:  $b_i = 0.4$ ,  $\tau = 0.3$ , Chatbot AI is used.

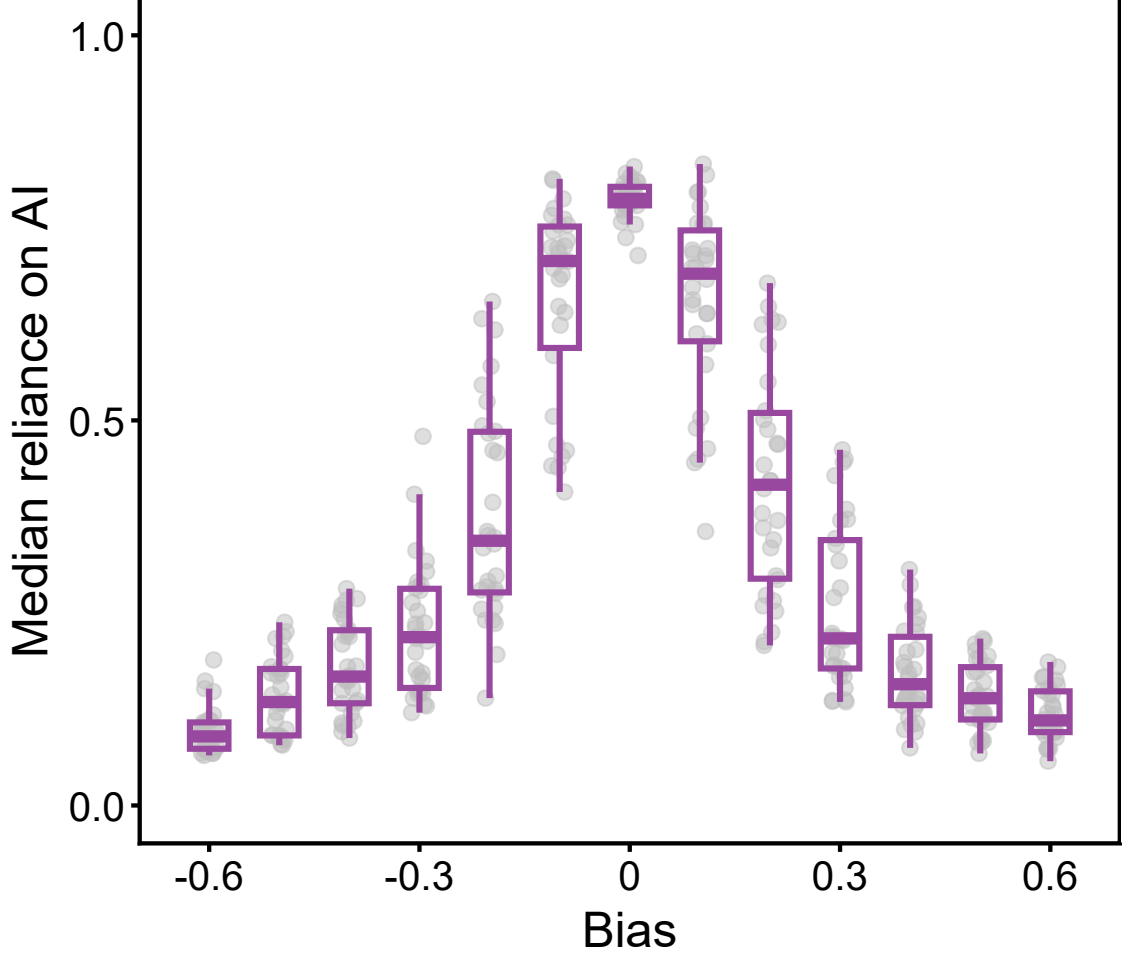


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 5^2)$ , and  $\beta_A \sim U(0, 1)$ . AI parameters:  $\tau = 0.3$ , Chatbot AI is used.

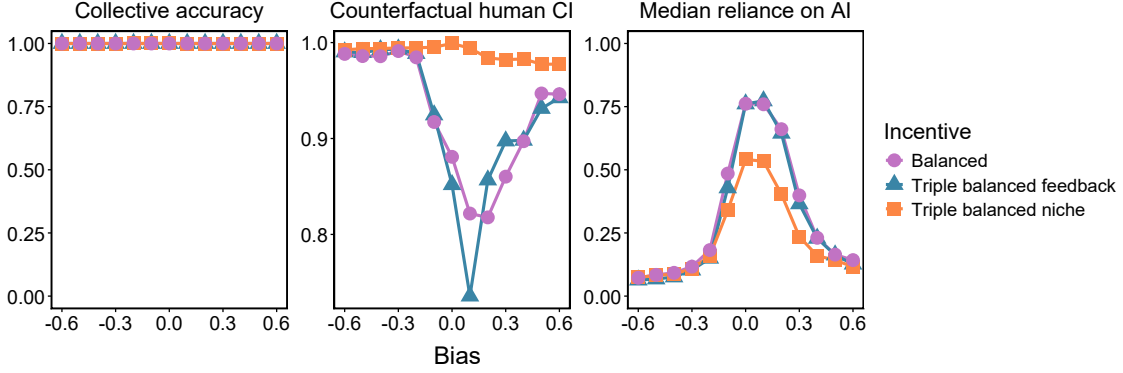


Figure S9: **Incentive structures incorporating unrevised prediction accuracy can promote better counterfactual human CI.** Evaluation metrics under the balanced, triple balanced feedback, and triple balanced niche structures across varying levels of AI bias. The triple balanced feedback structure is defined as the arithmetic mean of the feedback, niche-expert, and feedback based on unrevised predictions, whereas the triple balanced niche structure is defined as the arithmetic mean of the feedback, niche-expert, and niche-expert based on unrevised predictions. A larger absolute AI bias indicates a less accurate AI model. All evaluation metrics are averaged over generations 190,001–200,000. The triple balanced niche structure maintains higher counterfactual human CI across AI bias levels than the other incentive structures. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 5^2)$ , and  $\beta_A \sim U(0, 1)$ . AI parameters:  $\tau = 0.3$ ; Chatbot AI is used.

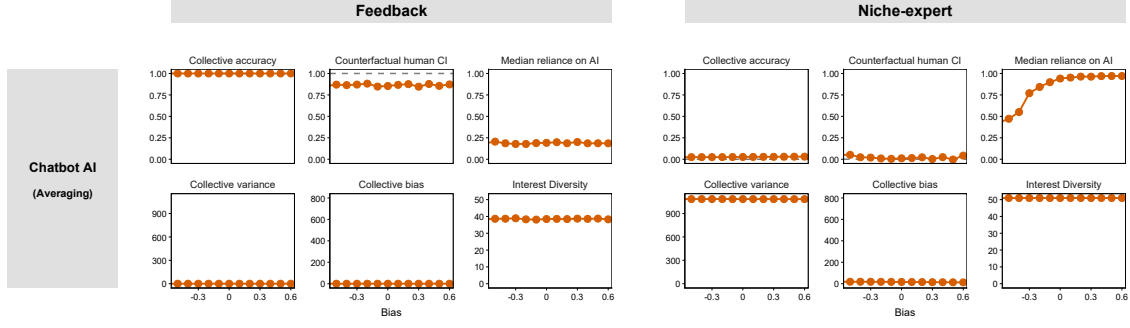


Figure S10: **Chatbot AI with averaging aggregation has a limited influence on CI.** Evaluation metrics across varying levels of AI bias under Chatbot AI with averaging aggregation. Two incentive structures, feedback and niche-expert, are considered. A larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, whereas counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as the Hill–Shannon diversity of interests within the population, and median reliance on AI ( $\beta$ ) across individuals. Collective bias represents the systematic deviation of the collective prediction from the true target, whereas collective variance represents the variability of collective prediction errors across prediction events. All evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters:  $M = 50$ ,  $s = 50$ ,  $\mu = 0$ ,  $\alpha_i \sim U(-5, 5)$ , and  $\sigma_i \sim U(0, 3)$ . Population parameters:  $N = 10,000$ ,  $c_A \sim \mathcal{N}(0, 100^2)$ , and  $\beta_A \sim U(0, 1)$ . AI parameters:  $\tau = 0.3$ .



Table S1: Evaluation metrics used to analyze collective prediction dynamics.

| Metric                             | Definition                                                                         | Interpretation                                                                                                         |
|------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Collective accuracy                | $1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$ | Accuracy of the revised collective prediction relative to the baseline prediction $\alpha_0$ .                         |
| Collective bias                    | $\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$                               | Squared expected deviation of the revised collective prediction from the true target.                                  |
| Collective variance                | $\text{Var}(Y - \hat{Y}^{\text{re}})$                                              | Variability of the revised collective prediction error across prediction events.                                       |
| Counterfactual collective accuracy | $1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$             | Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs. |
| Interest diversity                 | $\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$                                 | Effective number of interests represented in the population.                                                           |
| Median reliance on AI              | $\text{median}_{A \in N}(\beta_A)$                                                 | Median reliance on AI in the population.                                                                               |

## References

- Hong, L., S. E. Page, and M. Riolo (2012). Incentives, information, and emergent collective accuracy. *Managerial and Decision Economics* 33(5-6), 323–334.
- Mann, R. P. and D. Helbing (2017). Optimal incentives for collective intelligence. *Proceedings of the National Academy of Sciences* 114(20), 5077–5082.
- Taylor, P. D. and L. B. Jonker (1978). Evolutionary stable strategies and game dynamics. *Mathematical biosciences* 40(1-2), 145–156.
- Traulsen, A., J. C. Claussen, and C. Hauert (2005). Coevolutionary dynamics: from finite to infinite populations. *Physical review letters* 95(23), 238701.
- Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Individual incentives that promote collective intelligence. *Proceedings of the National Academy of Sciences* 122(51), e2516535122.